

TrialCheck

Product Requirements Document — v0.2.0

Platform-agnostic A/B experiment readout auditor

Field	Value
Status	Released — v0.2.0
Author	Sidharth Kriplani
Language	Python 3.10 / 3.11 / 3.12
License	MIT
Dependencies	zero (stdlib only)
Tests	17 unit tests, 3 test classes
Demo scenarios	4 canonical scenarios
PyPI	trialcheck

1. Problem Statement

Experimentation platforms surface a p-value and a lift estimate. That is not sufficient information to make a trustworthy ship decision. The gap between 'statistically significant result' and 'safe to ship' is filled by a checklist of questions that experienced practitioners know to ask — and less experienced ones do not.

The institutional cost of skipping this checklist is concrete:

- An SRM caused by a buggy assignment condition means the treatment and control groups are not comparable — the p-value is meaningless, but the team ships anyway.
- A result called at 30% of planned duration with repeated interim looks has an effective false-positive rate far above alpha — but no one computed it.
- A statistically significant lift of 0.02% on a metric with a 1% business threshold is not worth shipping — but the p-value says 'go'.
- A guardrail metric (refund rate, latency, error rate) moved in the wrong direction but wasn't checked because it wasn't the primary metric.

These failures are not caused by bad intentions. They are caused by the review being informal, inconsistent, and dependent on who happens to attend the readout.

2. Goals and Non-Goals

Goals:

- Package the senior DS readout checklist into a single zero-dependency library call
- Return per-check PASS/WARN/FAIL/INSUFFICIENT_INPUT with recommendations, not just verdicts
- Produce machine-parseable output (JSON) for CI gates alongside human-readable formats (MD, HTML)
- Cover the seven most common readout failure modes: SRM, peeking, practical significance, MDE context, statistical significance (binary + continuous), guardrail movement, pre-period imbalance
- Work with any experimentation platform's output — no platform-specific integration required

Non-Goals:

- Running or configuring experiments (TrialCheck is a readout auditor, not an experiment platform)
- CUPED / variance reduction (requires raw individual-level data; TrialCheck works on summaries)
- Multiple comparison correction across metrics (family definition is team-specific; v0 defers to user)
- Network effects, SUTVA, novelty effects — require domain knowledge and additional data

- Multi-variant ($k > 2$) SRM checks — scoped to two-variant tests in v0
- Automatic ship/no-ship decisions

3. Target Users

User	Use case	How they use the output
Data Scientist / Analyst	Audit readout before shipping; catch risks that prevent the experiment from shipping	Parse the output into JSON to experiment record
Analytics Lead / Reviewer	Standardise review process across team; require a trial check PASS/WARN	Create Trial Check Pass/WARN status in PR or experiment track
ML Platform / MLOps	Block experiment shipping in CI if FAIL; require a report over a status	Parse report over a status from JSON in pipeline
Interview candidate	Demonstrate systematic experimentation engineering practices	Show PDF and discuss statistical choices in interview

4. Check Specifications

Check	Required input	Stat method	FAIL condition	WARN condition
SRM	$n_{\text{control}}, n_{\text{treatment}}, \text{expected_control_ratio}$	χ^2 test	$p < \alpha$	—
Pre-period balance	PrePeriodCovariate list	$\text{SMD} = \Delta \text{mean} / \text{pooled_SD}$	—	Any $\text{SMD} \geq 0.1$
Primary metric	n , conversions per arm, alpha	Two-proportion z-test, pooled SE	—	—
Continuous metric	mean, std, n per arm, alpha	Welch t-test, W-S df	$p > \alpha$	INSUFFICIENT_INPUT if mean/std abs
Peeking risk	actual vs planned duration	Duration ratio heuristic	—	ratio < 1 OR interim_looks > 0
Practical significance	observed lift, practical_lift_threshold?	—	—	lift $<$ threshold
Guardrail movement	GuardrailMetric list: value, direction, tolerance	Directional tolerance check	Any metric breaches tolerance	Any metric breaches direction within tolerance

5. Input / Output Model

Input — ExperimentSummary:

Field	Type	Required	Description
experiment_id	str	Yes	Unique experiment identifier
metric_name	str	Yes	Name of the primary metric
control / treatment	VariantSummary	Yes	n, conversions, optional mean/std per arm
expected_control_ratio	float	Yes	Planned fraction assigned to control (e.g. 0.5)
alpha	float	Yes	Significance threshold (typically 0.05)
planned_duration_days	int	Yes	Planned test duration
actual_duration_days	int	Yes	Actual test duration at readout
interim_looks	int	Yes	Number of times results were checked before readout
mde	float	No	Minimum detectable effect the test was powered for
practical_threshold	float	No	Minimum lift considered business-meaningful
guardrails	list[GuardrailMetric]	No	Secondary metrics with bad direction and tolerance
pre_period_covariates	list[PrePeriodCovariate]	No	Pre-experiment covariate means/stds per arm

Output — TrialReport:

Field	Type	Description
overall_status	CheckStatus	PASS / WARN / FAIL / INSUFFICIENT_INPUT — worst across all checks
checks	list[CheckResult]	Per-check status, finding text, recommendation
interpretation	str	One-sentence plain-language summary
write_report(report, path)	None	Writes JSON / MD / HTML depending on file extension

6. Acceptance Criteria

The following must all hold before v0.2.0 is considered complete:

Criterion	Quantitative bar	Verified
SRM check fires on 56/44 assignment with 50/50 expected, status FAIL	50/50 expected, status FAIL	Yes

Clean experiment with full data returns PASS	Overall status PASS, 0 FAIL/WARN	Findings
Peeking WARN fires at 36% duration + 2 interim looks	Status WARN, check_name peeking	ask
Guardrail FAIL fires when revenue/user drops 4.4%	Stayer FWA tolerance	Yes
Welch t-test matches scipy.stats.ttest_ind equal_var=False	Basic validation	Yes
Zero pip dependencies — stdlib only	pip show trialcheck shows no Requires	Yes
17/17 unit tests pass on Python 3.10, 3.11, 3.12	CI matrix green	Yes
JSON output parseable with json.loads()	No serialisation errors	Yes
INSUFFICIENT_INPUT returned (not crash) when test_data absent	test_data absent missing_data_insufficient_input	Yes

7. Failure Scenarios That Drove Design Decisions

Scenario	What went wrong historically	How TrialCheck addresses it
SRM from sticky sessions	Users were bucketed by session, not user; SRM check fired; FAIL stops blocks shipping	SRM check changed to user
Peeking at interim results	Team checked results daily; stopped at first peeking 0.05; cache; WARN 0.30h recommendation to extend	Peeking 0.05; cache; WARN 0.30h
Significant but negligible lift	CTR +0.003pp, p=0.01; shipped; no reversal	Practical significance check; WARN when lift < threshold
Guardrail regression buried	Check found improved, error rate doubled	Guardrail primary metric check; FAIL on tolerance breach
Pre-period imbalance undetected	Active users over-represented in treatment; Pre-period imbalance check; WARN on SMD ≥ 0.1	Pre-period imbalance check; WARN on SMD ≥ 0.1

8. Summary

Dimension	v0.2.0 Status
All 7 checks implemented	Done
17 unit tests passing	Done
4 canonical demo scenarios	Done
Zero dependencies	Done
CI matrix Python 3.10/3.11/3.12	Done
JSON + MD + HTML output	Done
Explicit claim boundary in every report	Done
PyPI OIDC publish workflow	Done

Resume-safe claim: Built TrialCheck, a platform-agnostic A/B experiment readout auditor that checks completed experiment summaries for sample-ratio mismatch (chi-square), peeking risk, MDE context, practical and statistical significance (two-proportion z-test and Welch's t-test), guardrail movement, and pre-period covariate imbalance (SMD), producing JSON/Markdown/HTML audit reports with explicit decision caveats. Zero dependencies. 17 tests. 4 canonical demo scenarios.